

Software Architecture and Design Specification

Project: Voice Assistant for Multimedia Controls

Authors: Team 3

Dhanush S S: PES2UG24CS154

G S Dheekshith: PES2UG24CS166

Harsh Mannan: PES2UG24CS183,

Chiranthan Shankar: : PES2UG24CS137

Revision History

Version	Date	Author	Change Summary
1.0	13-09-2026	Team 3	Initial Software Architecture and Design Specification, derived from SRS v1.1 and the Team 3 Software Test Plan v1.1.

Approvals

Role	Name	Signature/Date
Course Coordinator / Instructor	TBD	Pending
Team Lead / Project Manager	Dhanush S S	Pending

1. Introduction

1.1 Purpose

This document specifies the software architecture and design of the Voice Assistant for Multimedia Controls, an ESP32-based hardware device that lets a user control media playback on a paired host device (phone, PC, TV, or Bluetooth speaker) using voice commands. It translates the functional, non-functional, and security requirements in the Team 3 Software Requirements Specification (SRS v1.1) into a concrete module architecture, component interfaces, and design decisions that guide implementation and verification against the Team 3 Software Test Plan (STP v1.1).

1.2 Scope

Covers the on-device firmware architecture (wake-word detection, voice capture, cloud ASR communication, command mapping, BLE HID transmission, connectivity/pairing, diagnostics, and OTA update), the external interfaces to the Cloud ASR service and the paired host device, and the optional PC-side Python + MediaPipe companion module. It does not cover the internal implementation of the host media player or any streaming-service integration, which remain out of scope per the SRS.

1.3 Audience

Developers (Team 3 module owners), QA/cross-testers, the Course Instructor and Project Evaluators, and future maintainers of the firmware.

1.4 Definitions

ESP32, ASR, BLE, HID, MCU, OTA, API, UI, SNR, TLS, Wake Word, RTM. Full definitions are provided in Section 1.4 of the SRS and are reused unchanged in this document.

2. Document Overview

2.1 How to use this document

This document provides the architectural deliverables for the project: a component/module diagram, module descriptions, the chosen architecture pattern and rationale, technology stack, risk register, requirements traceability, a STRIDE-based security architecture, two UML sequence diagrams covering the two flows most central to the project, API/interface definitions for the key module boundaries, error handling and monitoring conventions, and UX design notes.

2.2 Related Documents

- Software Requirements Specification (SRS) — Voice Assistant for Multimedia Controls, Version 1.1, Team 3.
- Software Test Plan (STP) — Voice Assistant for Multimedia Controls, Version 1.1, Team 3.
- Requirements Traceability Matrix (RTM) — Section 8 of the SRS and Section 15 of the STP.

3. Architecture

3.1 Goals & Constraints

Goals: reliable end-to-end voice-to-action latency ($\leq 3s$ for 90% of commands), $\geq 90\%$ command recognition accuracy at $SNR \geq 15dB$, secure audio transport and BLE pairing, and a modular firmware structure that lets each of the four owned modules be developed, cross-tested, and updated independently.

Constraints: ESP32 RAM/flash limits on-device processing to wake-word detection only, so full transcription must run in the cloud; BLE range is $\sim 10m$; the host device must support the BLE HID media-key profile; cloud ASR usage requires a stable Wi-Fi connection and is subject to third-party API rate limits/cost.

3.2 Stakeholders & Concerns

Stakeholder	Primary Concerns
End User	Fast, hands-free, reliable media control; clear feedback; privacy of voice data.
Developer / Maintainer	Modularity, ease of flashing/configuring, diagnostic visibility, safe OTA updates.
Course Evaluator	Requirements coverage, traceability, and testability against the SRS/STP.
Security reviewer	TLS-protected audio transport, authenticated BLE pairing, no plaintext credential or raw-audio retention, signed OTA images.

3.3 Component (UML) Diagram

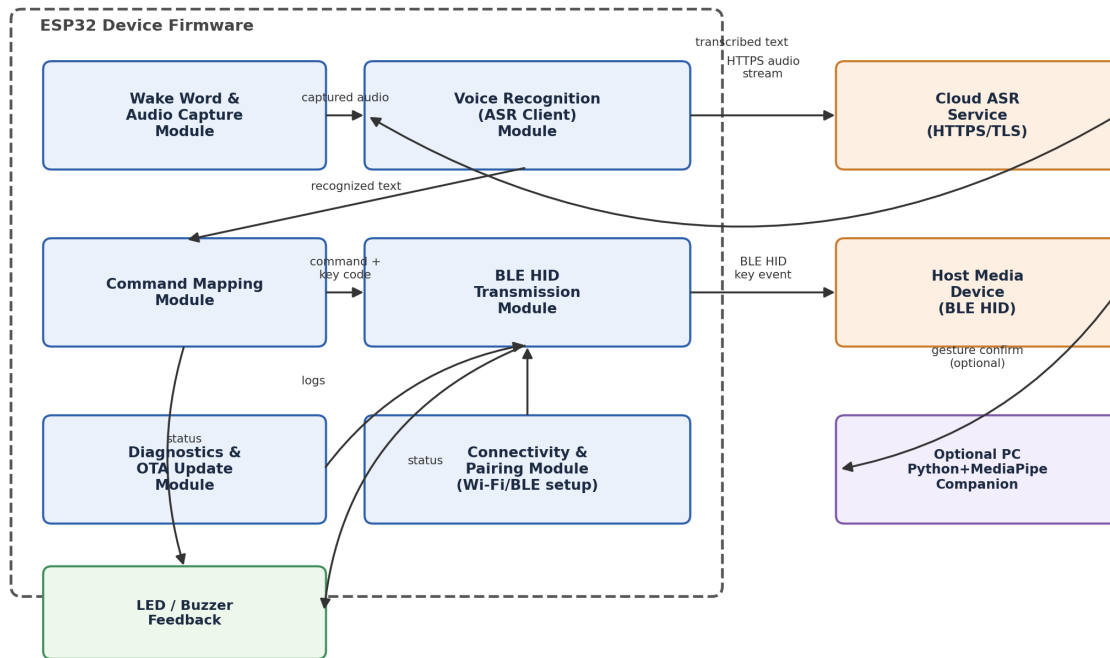


Figure 1: Component / Module Architecture — Voice Assistant for Multimedia Controls

3.4 Component Descriptions

- Wake Word & Audio Capture Module — continuously monitors the microphone on-device for the configured wake word and, once triggered, buffers up to 5 seconds of command audio; also owns the physical mute/unmute control (VAMC-F-001 to F-003).
- Voice Recognition (ASR Client) Module — streams captured audio to the Cloud ASR service over HTTPS/TLS, receives the transcription, and matches it against the 7-command vocabulary, flagging unrecognized/ambiguous input (VAMC-F-004 to F-006).
- Command Mapping Module — translates a recognized command into the corresponding BLE HID media-key code (VAMC-F-007).
- BLE HID Transmission Module — transmits the key event to the paired host within 2 seconds, with retry/backoff (up to 3 attempts) on transient failures, and drives success/error feedback (VAMC-F-008 to F-010).
- Connectivity & Pairing Module — handles first-boot Wi-Fi provisioning (captive portal), Bluetooth pairing, and factory reset (VAMC-F-011 to F-013).
- Diagnostics & OTA Update Module — writes recognized commands/errors to a local circular log buffer retrievable via serial console, and applies signed OTA firmware images (VAMC-F-014, F-015, SR-005).
- LED/Buzzer Feedback — shared output used by multiple modules to indicate listening, recognized, success, error, and low-battery states.
- Optional PC Python + MediaPipe Companion — a development-time/testing aid for gesture-based command confirmation; not required for the core end-user experience.
- External actors — Cloud ASR Service (speech-to-text over HTTPS) and Host Media Device (BLE HID receiver that executes the media action).

3.5 Chosen Architecture Pattern and Rationale

A layered, event-driven embedded architecture was chosen: a Sensing/Capture layer (wake word, mic buffering), a Recognition layer (cloud ASR client, command matching), an Execution layer (command mapping, BLE HID transmission), and a Platform/Support layer (connectivity, diagnostics, OTA) that all layers depend on. Each layer maps to one team member's owned module (SRS Section 9.2), which keeps ownership boundaries and the physical module boundaries identical — directly supporting the modularity NFR (VAMC-NF-005) and the team's no-self-testing cross-testing rule.

A full on-device (offline) recognition architecture was considered and rejected: the ESP32's limited RAM/flash cannot host a full-vocabulary ASR model, and the SRS explicitly scopes recognition to a hybrid on-device-wake-word / cloud-transcription model. A microservices-style backend was not applicable, since there is no server-side application beyond the third-party Cloud ASR API and (optionally) the OTA distribution endpoint.

3.6 Technology Stack & Data Stores

- Hardware: ESP32-WROOM (or equivalent) with I2S/analog MEMS microphone, integrated 2.4GHz Wi-Fi and Bluetooth 4.2+/BLE radio, status LED, buzzer, mute/reset push-button, USB or 2000mAh Li-ion power.
- Firmware: C/C++ on the ESP-IDF / Arduino-ESP32 framework, structured as independent modules per Section 3.4.
- Cloud service: third-party Cloud Speech-to-Text API (e.g., Google Cloud Speech-to-Text) reached over HTTPS with TLS 1.2+.
- Host-side interface: standard BLE HID media-key profile on the host OS (Android/Windows/iOS/macOS) — no custom host-side software required.
- Optional companion: Python + MediaPipe on a PC, used only for development-time gesture testing.
- Data stores: no persistent database. Non-volatile flash stores Wi-Fi/BLE credentials (encrypted/obfuscated, never plaintext) and a bounded circular diagnostic-log buffer; no raw audio is persisted beyond an active recognition session.

3.7 Risks & Mitigations

Risk	Mitigation
Cloud ASR unavailable or rate-limited (no internet / API quota)	Graceful degradation: LED/buzzer error indication; retry/backoff for transient Wi-Fi failures; no offline fallback is in scope, so this is surfaced to the user rather than silently failing.
BLE link drops during command transmission	Retry/backoff up to 3 attempts (VAMC-F-009); error feedback via LED/buzzer if all retries fail.
False wake-word triggers or misrecognized commands	On-device wake-word threshold tuning; unrecognized/ambiguous commands are detected and rejected without executing unintended actions (VAMC-F-006).
Malicious or corrupted OTA image	OTA images must be signed; the bootloader verifies the signature before applying the update and rejects unsigned/tampered images (VAMC-SR-005).
Eavesdropping on Wi-Fi credentials or ASR audio in transit	TLS 1.2+ enforced for all cloud ASR traffic; credentials are never stored in plaintext in flash.
ESP32 RAM/flash limits constrain feature growth	Keep on-device logic limited to wake-word detection and thin clients; push transcription and heavier processing to the cloud.

3.8 Traceability to Requirements

Requirement(s)	Architectural Element
VAMC-F-001 to F-003 (wake word, capture, mute)	Wake Word & Audio Capture Module
VAMC-F-004 to F-006 (ASR streaming, matching, unrecognized handling)	Voice Recognition (ASR Client) Module + Cloud ASR external interface
VAMC-F-007 to F-010 (command set, BLE transmission, retry, feedback)	Command Mapping Module + BLE HID Transmission Module + LED/Buzzer Feedback
VAMC-F-011 to F-013 (Wi-Fi provisioning, pairing, factory reset)	Connectivity & Pairing Module
VAMC-F-014, F-015 (diagnostic logging, OTA)	Diagnostics & OTA Update Module
VAMC-NF-001 to NF-005 (latency, accuracy, usability, battery, modularity)	End-to-end pipeline design (Fig. 1–2); layered architecture (Section 3.5)
VAMC-SR-001 to SR-005 (TLS, authenticated pairing, credential/audio protection, signed OTA)	Security Architecture (Section 3.9); Diagnostics & OTA Update Module; Connectivity Module

3.9 Security Architecture

Threat modeling (STRIDE):

- Spoofing — BLE pairing uses authenticated pairing (passkey/numeric comparison) rather than "Just Works" where the host supports it (VAMC-SR-002), preventing unauthorized devices from impersonating the assistant or the host.
- Tampering — OTA firmware images must be signed and are verified by the bootloader before being applied; unsigned/tampered images are rejected (VAMC-SR-005).
- Repudiation — recognized commands and errors are logged to a local circular diagnostic buffer retrievable via serial console, supporting after-the-fact review (VAMC-F-014).
- Information Disclosure — all audio sent to the Cloud ASR service uses TLS 1.2+ (VAMC-SR-001); Wi-Fi credentials are not stored in plaintext in flash (VAMC-SR-003); raw audio is not retained beyond the active recognition session (VAMC-SR-004).
- Denial of Service — retry/backoff (up to 3 attempts) bounds reconnection attempts on transient BLE/Wi-Fi failures rather than retrying indefinitely; graceful LED/buzzer degradation avoids the device hanging when the network or Bluetooth link is unavailable.
- Elevation of Privilege — device configuration actions (Wi-Fi provisioning, pairing, factory reset, OTA) require physical access (button press) or a signed image, so a remote attacker cannot silently reconfigure the device.

4. Design

4.1 Design Overview

The firmware is organized into the six modules described in Section 3.4, communicating through well-defined internal function calls/event queues on-device and two external interfaces (Cloud ASR over HTTPS, and the host device over BLE HID). This keeps each module independently testable — matching the SRS's no-self-testing cross-testing plan — and independently upgradable, satisfying the modularity NFR (VAMC-NF-005).

4.2 UML Sequence Diagrams

Two sequence diagrams are provided, covering the two flows central to the project: the primary voice-control path, and the OTA/factory-reset maintenance path.

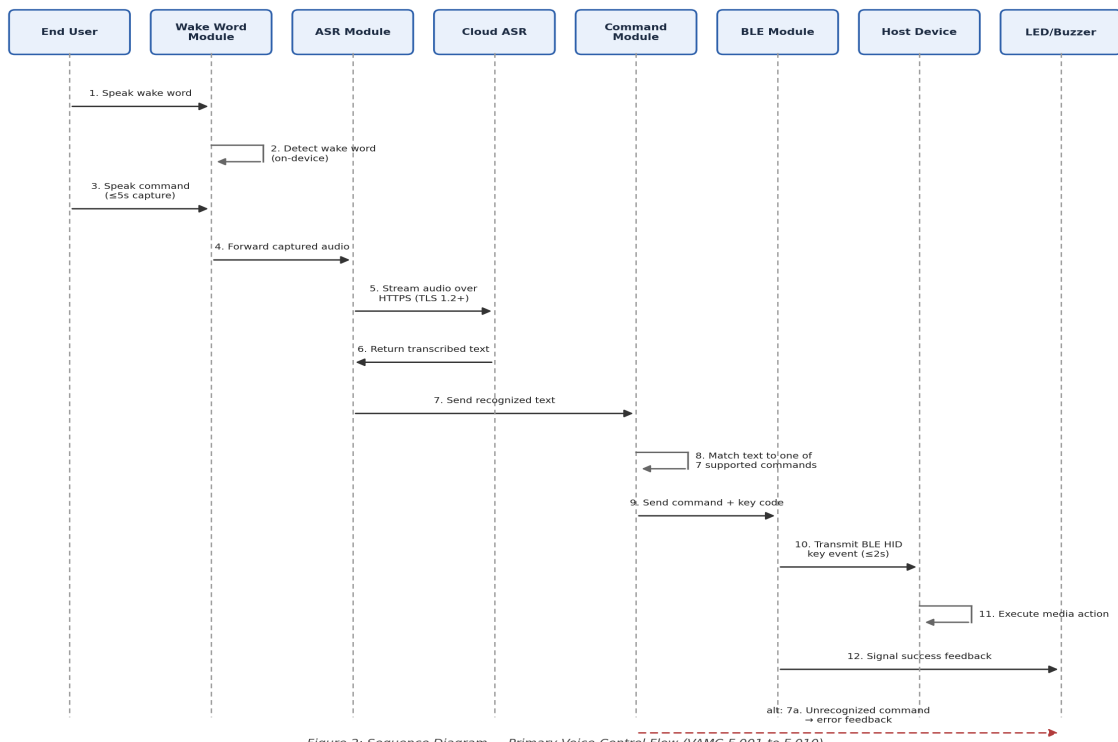


Figure 2: Sequence Diagram — Primary Voice-Control Flow (VAMC-F-001 to F-010)

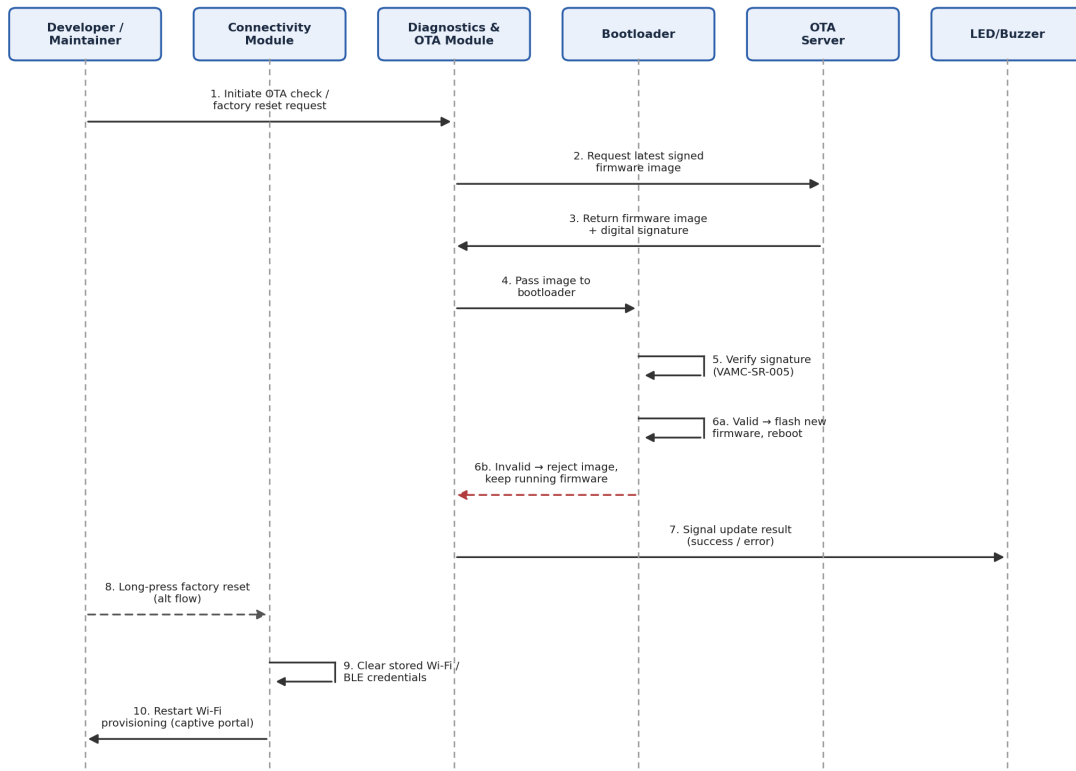


Figure 3: Sequence Diagram — OTA Firmware Update & Factory Reset Flow (VAMC-F-013, F-015, SR-005)

4.3 API Design

Interface definitions are provided for the two component boundaries with the highest integration risk: the ASR Module's external call to the Cloud ASR service, and the internal Command Mapping Module → BLE HID Transmission Module interface.

Interface 1: ASR Module → Cloud ASR Service

Endpoint: POST <https://speech.googleapis.com/v1/speech:recognize> (or equivalent Cloud ASR endpoint)

Transport: HTTPS, TLS 1.2+ (VAMC-SR-001)

Request: { config: { encoding: "LINEAR16", sampleRateHertz: 16000, languageCode: "en-US" }, audio: { content: <base64 audio> } }

Response: { results: [{ alternatives: [{ transcript: <string>, confidence: <float> }] }] }

Errors: 401 Unauthorized (invalid/expired API key), 429 Too Many Requests (rate limit), 5xx Service Unavailable (mapped internally to VAMC-F-006 unrecognized-command handling and LED/buzzer error feedback).

Interface 2: Command Mapping Module → BLE HID Transmission Module

Call: ble_send_hid_key(command_id, retry_count)

Request: { command: VOLUME_UP | VOLUME_DOWN | PAUSE | RESUME | PLAY | NEXT | PREVIOUS, timestamp }

Response: { status: SUCCESS | RETRIED | FAILED, latency_ms }

Errors: BLE_NOT_CONNECTED (host not paired/in range), BLE_TIMEOUT (no ack within 2s budget — triggers retry/backoff up to 3 attempts per VAMC-F-009).

4.4 Error Handling, Logging & Monitoring

Standardized error/status codes are used across module boundaries (see Section 4.3). No sensitive information (raw audio, Wi-Fi credentials) is written to logs; only recognized command IDs, timestamps, and error codes are logged to the local circular buffer (VAMC-F-014), retrievable via serial console for developer diagnostics. Monitoring during testing focuses on the metrics defined in the STP: end-to-end latency (90th percentile over 50 trials), recognition accuracy over 100 utterances, BLE transmission success/retry rate, and battery runtime in idle-listening mode.

4.5 UX Design

There is no onboard display. First-time setup uses a captive-portal Wi-Fi configuration page (or lightweight companion app) for Wi-Fi credentials and Bluetooth pairing, targeting completion within 5 minutes for a first-time user (VAMC-NF-003). A single multi-color status LED and buzzer communicate listening, recognized, success, error, and low-battery states, since the device has no other display. Setup and command feedback are designed to require no prior technical knowledge, consistent with the End User persona in the SRS (Section 2.3).

4.6 Open Issues & Next Steps

- Confirm final choice of Cloud ASR provider and finalize API credential provisioning/rotation strategy.
- Decide whether an IR transmitter path is implemented alongside BLE HID for hosts that lack BLE HID support.
- Define the exact encoding/obfuscation scheme used to protect Wi-Fi credentials in flash (VAMC-SR-003).
- Finalize the OTA distribution mechanism (hosting of signed images) referenced in Figure 3.